Soil Science

"The science dealing with soil as a natural resource on the surface of the earth, including Pedology (soil genesis, classification and mapping), physical, chemical, biological and fertility properties of soil and these properties in relation to their management for crop production."

Views on Soil (Science)

The term SOIL was derived from the Latin Word "SOLUM" Means FLOOR

- For a Layman soil is dirt or debris
- For an Agriculturist soil is a habitat for plant growth (to grow crops)
- For a Mining Engineer soil is a debris covering the Rocks
- For a Civil Engineer soil is a material on which road bed or house bed is formed
- For a Home Owner soil is a mellow or loamy or hard material

Definitions

Generally soil refers to the loose surface of the earth as identified from the original rocks and minerals from which it is derived through weathering process.

Whitney (1892): Soil is a nutrient bin which supplies all the nutrients required for plant growth

Hilgard (1892): Soil is more or less a loose and friable material in which plants, by means of their roots, find a foothold for nourishment as well as for other conditions of growth"

Dokuchaiev (1900): Russian scientist - Father of soil science - Soil is a natural body composed of mineral and organic constituents, having a definite genesis and a distinct nature of its own.

Joffe (1936): "Soil is a natural body of mineral and organic constituents differentiated into horizons - usually unconsolidated - of variable depth which differs among themselves as well as from the underlying parent material in morphology, physical makeup, chemical properties and composition and biological characteristics".

Jenny (1941): Soil is a naturally occurring body that has been formed due to combined influence of climate and living organisms acting on parent material as conditioned by relief over a period of time.

Ruffin and Simonson (1968): Soil is a mixture of Earth's uppermost mantle of weathered rock and organic matter

Buckman and Brady (1969): Soil is a dynamic natural body on the surface of the earth in which plants grow, composed of mineral and organic materials and living forms

Soil Science Society of America (1970)

(i) Soil is the unconsolidated mineral matter on the surface of the earth that has been subjected to and influenced by genetic and environmental factors of parent material, climate (including moisture and temperature effects), macro and microorganisms and topography, all affecting over a period of time and producing a product, that is "SOIL" that differs from the material from which it is derived in many, physical, chemical, biological and morphological properties and characteristics.

(ii) The unconsolidated mineral material on the immediate surface of the earth that serves as a natural medium for the growth of land plants.

Dr. W.E.H. Blum

Soils not only serve for agriculture and forestry, but also for filtering, buffering and transformation activities between the atmosphere and the ground water, protecting the food chain and drinking water against pollution and biodiversity

As soil provides nutrients, water, air and anchorage and supports life on Earth, it can be called as Soul Of Infinite Life (SOIL)

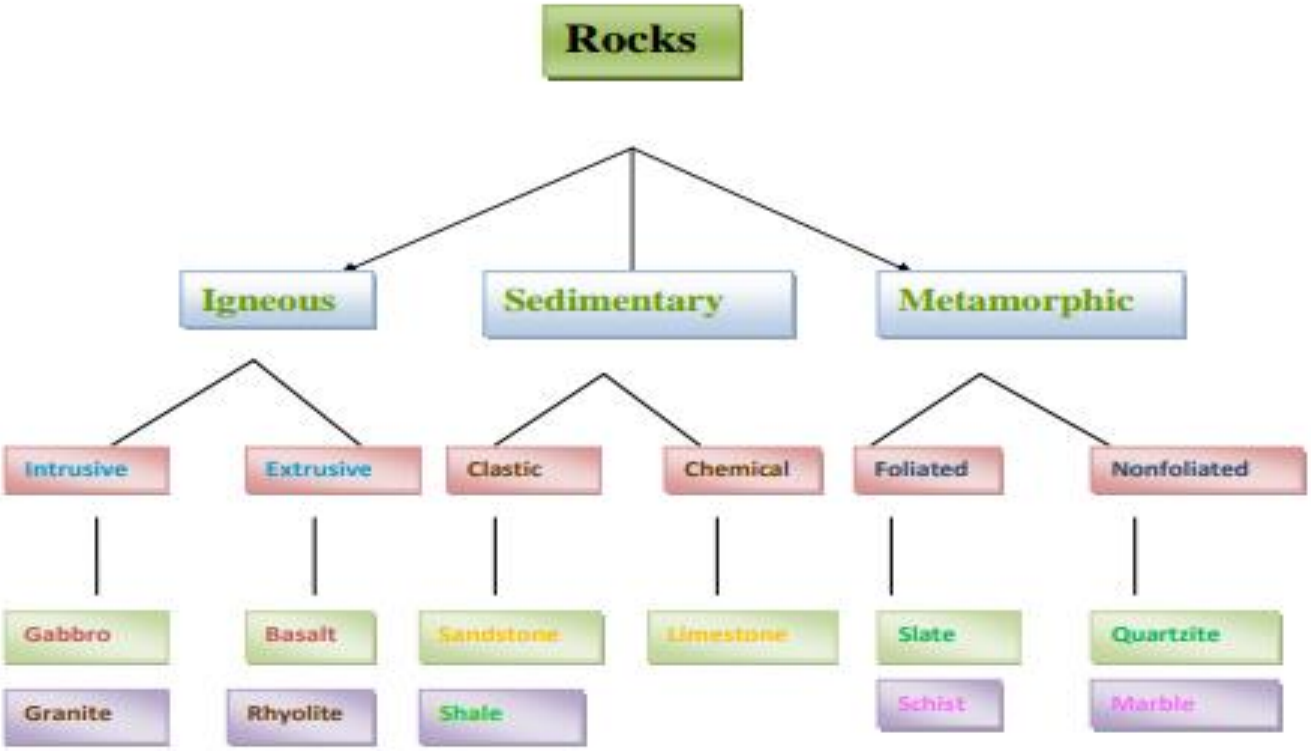
Rocks and minerals

What is Rocks?

Rocks are the materials that form the essential part of the Earth's solid crust. "Rocks are hard mass of mineral matter comprising one or more rock forming minerals". Rocks are formed from the molten material known as **magma**. The study of rocks is called **Petrology** (in Greek, petra means rock, logos means science). Petrology deals with the description of rocks; petrogenesis is the study of the origin of rocks.

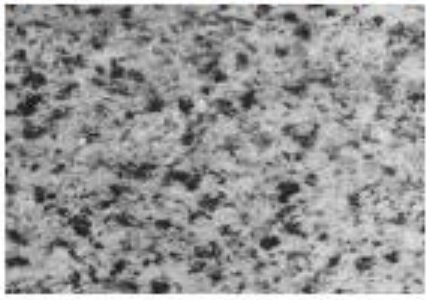
Formation of rocks

1. Cooling and consolidation of molten magma within or on the surface of earth = Igneous or **Primary rocks**
2. Transportation and cementation of primary rocks = **Sedimentary or Secondary rocks**
3. Alteration of the existing primary and secondary rocks = **Metamorphic rocks**



1. Igneous rocks (primary or massive rocks)

These are first formed in the earth crust due to the solidification of molten magma. Based on the mode of formation, they are further classified as extrusive and intrusive rocks



Extrusive rocks or volcanic rocks

These rocks are formed due to the consolidation of magma on the surface of the earth. The magma, when it flows on the Earth surface is called LAVA. E.g. Basalt.

Intrusive rocks or plutonic rocks

These rocks are produced due to solidification of magma below the surface of the earth. Plutonic – intrusive rocks solidifies at greater depth and Hypabassal rocks solidifies at shallow depth from the surface. E.g. Granite, syenite, diorite, Gabbro *etc.* Rocks formed in vertical cracks are called dykes and in horizontal cracks are called sills.

Vesicular rocks: Molten magma cools on the surface. Steam of water is entrapped into rocks and forms vesicles.

Based on the silica content, rocks are also classified as

1. Acid rocks : >65% SiO_2 (Granite, Rhyolite)
2. Intermediate : 56 to 65% SiO_2
   (Sub acid rocks 60 to 65% SiO_2 (Syenite and Trachyte))
   (Sub basic rocks 56 to 60 % SiO_2 (Diorite and Andesite))
3. Basic rocks : 40 to 55% (Gabbro, basalt)

Igneous rocks

S.No	Rocks	Origin	Essential minerals	Common minerals	Average specific gravity	Remarks
i.	Granite	Plutonic holocrystalline	Quartz (20 to 30%)	Hornblende, magnetite, mica	2.64	Light coloured white or reddish
ii.	Syenite	Plutonic Holocrystalline	Quartz, orthoclase	Hornblende, magnetite, biotite	2.80	Light coloured white or reddish
iii.	Diorite	Plutonic Holocrystalline	Quartz	Hornblende, magnetite, biotite	2.85	Darker
iv	Gabbro	Plutonic Holocrystalline	Labradorite, augite, olivine	Hornblende, ilmenite	3.0	Blakish

v.	Dolerite	Hypabasal	Labradorite, augite, olivine	Hornblende, ilmenite	3.0	Blakish
vi.	Basalt	Volcanic crystalline with glassy mass	Labradorite, augite, olivine	Hornblende, ilmenite	3.0	

2. Sedimentary rocks

These rocks are formed from the consolidation of sediments accumulated through wind or water action at the surface of the earth. Many are deposited in layer or formed through chemical reactions as precipitates from aqueous solutions. Sediments may contain various size particles cemented together by substances like SiO_2, Fe_2O_3 or lime. These rocks are also called as clastic rocks.

Based on the origin, the sedimentary rocks are classified as

1. Residual : Laterite
2. Transported
   - a. Deposited as solids in suspension : Sandstone, shale
   - b. Deposited by chemical precipitation : Limestone, ironstone
   - c. Deposited through agency of organic matter: Peat, Phosphatic deposits

Based on the grain size, sedimentary rocks are classified as

1. Rocks with boulder pebbles sized minerals (Rudaceous) : Conglomerate
2. Rocks with sand size particles (Arenaceous) : Sandstone
3. Rocks with silt size particles (silt rocks) : Siltstone
4. Rocks with clay size particles (Argillaceous) : Shale

Sedimentary rocks

S.No	Rock	Mineral composition	Colour and structure
1.	Sandstone	Mainly quartz with some $CaCO_3$, iron oxides and clay	Light to red, granular
2.	Shale	Clay minerals, quartz and some organic matter	Light to dark thinly laminated
3.	Limestone	Mainly calcite with some dolomite, iron oxides, clay, phosphate and organic matter	Light grey to yellow, fine grained and compact



3. Metamorphic rocks

These are formed from igneous and sedimentary rocks under the influence of heat, pressure, chemically active liquids and gases. Change may occur in mineral composition or texture or both. The changes due to water is called **hydro metamorphosis** and due to pressure is called **dynamo metamorphosis**

Sand stone : Quartizite

Shale : Slate/mica, schist

Lime stone : Marble

Granite : granite gneiss

Dolerite : Hornblende gneiss



Metamorphic rocks

S.No.	Rock	Mineral composition	Colour and structure
1.	Gneiss	Formed from granite	Alternating light and dark colours, banded and foliated
2.	Schist	Formed from basalt or shale	As original rock, foliated
3.	Quartzite	Formed from	Light or brown, compact and uniform texture,

		sandstone	foliated structure
4.	Slate	Formed from shale	Grey to black, compact and uniform texture, foliated structure
5.	Marble	Formed from lime stone	Light red, green, black, compact fine to coarse texture, foliated structure

Brief description of important rocks-Mineralogical composition

Sedimentary rocks

Formed through the agency of water. Also called as **aqueous rocks**. Formed from sediments brought by water. The sediment may contain various types of substances and sizes of particles. The particles are cemented by silica, iron oxide or lime to give a consolidated form. The rocks are mostly deposited in layers or strata – so called as **stratified rocks**. Sometimes they are formed by cooling, evaporation or by direct chemical precipitation. Anyway they are of secondary origin.



Sedimentary rocks divided into six groups as follows

1. **Arenaceous:** Formed of the deposits of coarse grained particles. They are composed of siliceous material derived from the disintegration of older rocks. The fragmental material so derived is deposited in beds of varying thickness through the agency of water. Depending upon the nature of cementing material present, some arenaceous rocks are hard and refractory, but most are loose and fall away very easily. E.g. Sandstone, grit, conglomerate and breccia.
2. **Argillaceous rocks:** Consist of small sized particles known as clay. They are composed of hydrated silica of alumina in admixture with sand, various other silicates and calcareous matter. When clay is deposited mainly of silicate of alumina, it is known as kaolin or China clay. E.g. clay, mudstone, shale and fuller's earth.

3. **Calcareous rocks:** Consists of carbonate of lime or lime and magnesia. They may be of sedimentary origin or formed by chemical precipitation or by organic agency. When, they are of organic agency, they are composed mainly of debris from plant and animal life. They are formed either by growth and decay of organisms in situ or by the transport and subsequent accumulation of their remains. The rocks so formed are found in layers, which vary considerably in depth of thickness.

When formed by chemical precipitation, the calcareous material is deposited in the form of layers/sheets from waters containing calcium carbonate in solution. The precipitate when first formed is usually soft and chalky, but soon acquires a hard, compact structure and crystalline texture. The important calcareous rocks of aqueous origin are limestone, chalk, magnesian, ferruginous limestones, dolomite, marks of various varieties and coral.

4. **Carbonaceous rocks:** Formed from decomposing vegetation under anaerobic conditions. When plants undergo decomposition under restricted air supply, is greater portion of the carbonaceous matter is retained and the material is slowly converted into coal. E.g. peat, lignite, coal, anthracite.

5. **Siliceous rocks:** Siliceous rocks of organic origin formed from parts of minute plants and animals like diatoms, radiolaria etc, some are soft and friable and crumble to powder very easily. Others like flint and chert are hard and compact.

6. **Precipitated salts:** Consist mainly of deposits formed as rock masses either by cooling, evaporation or by chemical precipitation. Water charged with acid or alkaline material, acting under pressure as it does under subterranean regions, dissolves various mineral substances from rocks with which it comes in contact. The salts thus formed deposit as rocks and such rocks vary in composition. They are

   i. Oxides: e.g. hematite, limonite, bauxite and quartz.
   ii. Carbonates: e.g. stalactite, stalagmite, magnetite and limestone.
   iii. Sulphates: e.g. gypsum and anhydrite
   iv. Phosphates: e.g. phosphorite
   v. Chlorides: e.g. rock salt.

Metamorphic rocks

The igneous and sedimentary rocks after they were first formed sometimes undergo a change. When the change is considerable, the rock is said to have undergone metamorphosis and the new rock is known as a metamorphic rock. The metamorphism is brought about by the action of water, heat or pressure or by the combined action of any one of these or all. The change brought about by water is hydro-metamorphism. The change brought about by heat is thermo-metamorphism. The change brought about by pressure is

dynamo- metamorphism. The changes that are brought about are both physical and chemical in character. In some cases the metamorphism is so pronounced that the new rock looks quite different from the original.

The action of water tends to remove some material or introduce new materials. By the introduction of a cementing material like silica, lime or iron oxide, loose sand may be turned into sandstone or sandstone into a quartzite. By the removal of certain constituents by percolating waters, basalt or granite may be converted into a laterite.

The action of heat hardens the rock and develops new crystals in it. Crystalline marble is produced this way from amorphous limestone by the action of heat and pressure. Due to pressure, the crystals of the original rock get pressed or flattened and the new rock is foliated. When foliation is slight, the layers are inseparable and it is called as gneiss. It foliation is complete with distinct and separable layers it is called as schist.

Mineralogical composition of important rocks

SI. No	Rocks	Grain size	Essential minerals	Most common accessory minerals	Average specific gravity	Remarks
(i)	Igneous rocks					
1.	Granite	Plutonic Holocrystalline	Predominant quartz 20-35 orthoclase	Hornblende mica, magnetite	2.64	Light coloured, White or reddish
2.	Syenite	- do -	Predominance quartz 100% plus of orthoclase, nepheline and albite	Hornblende, biotite, magnetite	2.08	- do -
3.	Granodiorite	- do -	Intermediate quartz plagioclase exceeds orthoclase	- do -	2.07	Medium coloured reddish
4.	Diorite	- do -	Intermediate quartz absent plagioclase	- do -	2.85	Darker
5.	Gabbro	- do -	Labradorite augite + olivine	Hornblende, ilmenite	3.00	Blackish
6.	Dolerite	Hypabyssal ophitic texture	- do -	- do -	3.00	- do – -

7.	Basalt	Volcanic micro crystalline with glassy mass	- do -	- do -	3.00	- do -

SI.No	Name of the type	Mineral composition	Colour and structure
(ii)	**Sedimentary rocks**		
1.	Sandstone	Mainly quartz with same contents, such as calcium carbonate, iron oxides and clays	Light to red, usually granular in structure
2.	Shale	Clay minerals, quartz and some organic matter	Light to dark thinly laminated structure
3.	Limestone	Mainly calcite or calcite and dolomite with some iron oxides, clay, phosphate and organic matter	Usually light grey to yellow usually fine grained and compact
(iii)	**Metamorphic rocks**		
1.	Gneiss	Formed from granite, mineral composition like that of granite	Alternating light and dark colours. Banded and foliated structure
2.	Sehist	Formed from basalt or shales. Mineral composition much as that of original rock	Much as original rock, foliated structure
3.	Quartzite	Formed from sandstone and of same composition	Light to brown. Compact and uniform texture, foliated structure
4.	Slate	Formed from shale and of same composition	Grey to black, compact, and uniform texture